

CLADISTICS

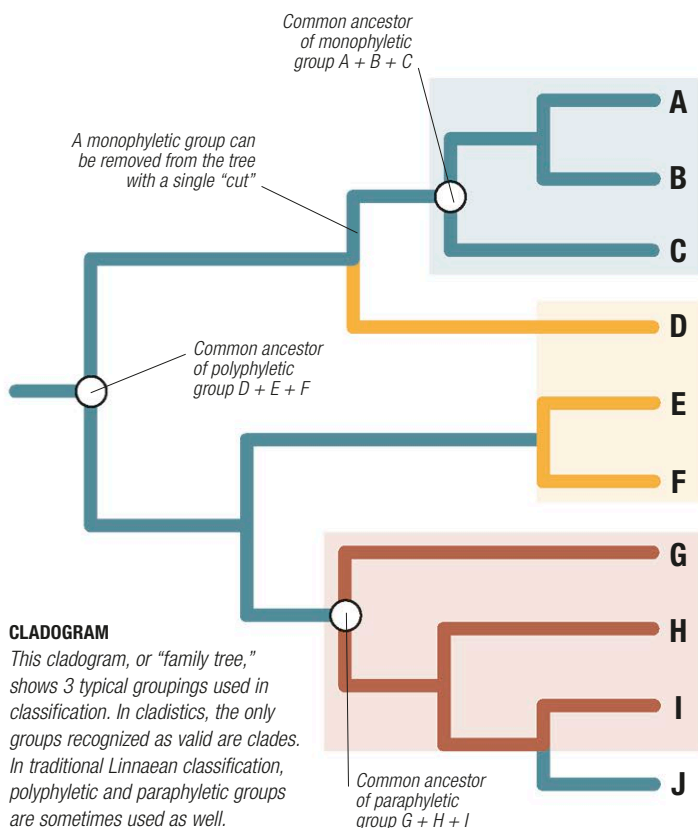
During the 1950s, a new classification system was proposed by the German entomologist Willi Hennig (1913–1976). Unlike traditional Linnaean classification, Hennig's system—called cladistics, or phylogenetics—works solely with hierarchical groups called clades. Clades can contain a handful of species, or hundreds of thousands, but each is a single branch from the tree of life, with an ancestor and all of its descendants. Cladistics was controversial when it was first introduced, but with the advance of DNA analysis in the 1990s, it has moved to the forefront of modern classification.

Clades

Cladistics investigates the sequence in which features have evolved. It makes 2 basic assumptions: that living things are all related—however distantly—and that features evolve over time. By taking a group of species, and then comparing the features that they do or do not share, taxonomists generate cladograms, or “family trees,” showing the most likely path that evolution has followed.

Cladistics differs from Linnaean classification in several ways. For example, the only groups cladistics recognizes are clades. These can be of any size, but by definition they are monophyletic,

meaning that they contain a complete evolutionary line, with all the offshoots that it has produced. Linnaean classification, on the other hand, long predates the discovery of evolution. Many of its groups are indeed clades, but some are polyphyletic, meaning that they contain several evolutionary lines treated as if they were one. Others are paraphyletic, which means that they contain some branches of a clade, but not all. For example, Linnaean classification recognizes reptiles as a separate class, and it gives the same treatment to birds. Cladists recognize them both as members of the same clade, which also includes the dinosaurs.



CLADE

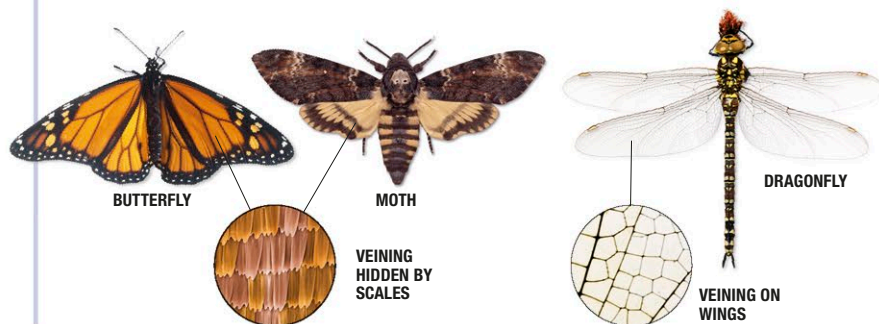
Also known as a monophyletic group, this contains a single shared ancestor, and all its descendants, both alive and extinct. A clade can be of any size, and may contain further clades nested within it.

POLYPHYLETIC GROUP

Unlike a clade, this contains a number of species, but not their shared ancestor. Warm-blooded animals, for example, form a group like this: they consist of mammals and birds, but do not form a clade.

PARAPHYLETIC GROUP

A paraphyletic group contains an ancestral species, but not all of its descendants. As a result, it does not form a clade. Reptiles are an example, because, traditionally, they do not include birds.

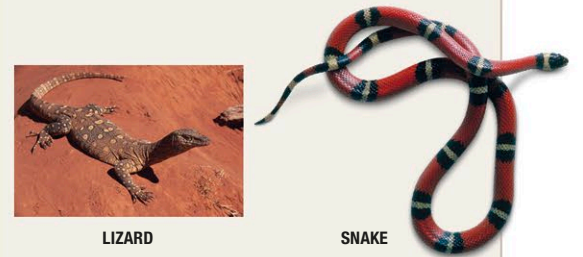


REVEALING WINGS

All flying insects have evolved from a single common ancestor that had 6–8 strengthening veins along its wings. In dragonflies, the primitive pattern of veins has been subdivided many times. But in butterflies and moths, it is usually hidden by scales; these scales point to their shared ancestry.

GENETIC INSIGHTS

When used with cladistics, genetic evidence often confirms traditional classification, but in some cases it refutes existing ideas. For example, snakes and lizards are traditionally classified in the order Squamata, which also includes the amphisbaenians; some snakes have the vestiges of pelvic bones, which strongly suggests that snakes had lizardlike ancestors. However, DNA evidence shows that the picture may be more complex. Snakes are now widely regarded as a monophyletic group, and constitute one of several major clades within the Squamata, with the other clades including lizards.



LOSING LEGS

Lizards typically have well developed legs, many rows of scales on their undersides, and eyelids that can blink. By contrast, snakes are legless, with a single row of belly scales, and eyes covered by a single transparent scale, or brille. These adaptations may have evolved more than once, originally for a burrowing lifestyle.

Cladistic analysis

When cladistics was first introduced, it relied on structural features, such as bones, feathers, or vein patterns on insects' wings. It still uses these today, but in addition, modern cladistics works by comparing genes. Whatever the features, the procedure is the same: the “primitive” (ancestral) or “derived” (altered) state of various features is scored and compared for a particular group of species. Those sharing the most derived features are likely to be most closely related—a principle that is used to build up the branch-points of a family tree. For example, moths and butterflies share numerous features with other insects, but they differ in being covered in microscopic scales. This derived feature—together with many others—strongly suggests that they are more closely related to each other than to other insects. Initially, cladists processed their data by hand, but today's computers can deal with thousands of features, analyzing huge numbers of outcomes to produce the most probable result.

Two systems

Linnaean classification has a specific hierarchy of named levels. Cladistics does not work like this, because branch-points can occur at any stage in a family tree. As a result, cladists prefer not to give clades different levels of their own. Instead, they are often referred to as “unranked clades,” followed by a definition or shortened name.

Taken to its logical conclusion, cladistics rewrites much of animal classification, creating a plethora of unranked clades in the place of traditional Linnaean groups. However, the Linnaean system is very practical, which is why the 2 systems are often used side by side. Cladistics provides a powerful way of investigating the past, and determining how closely different species are related, while the Linnaean system provides a concise way of identifying and dealing with familiar groups and their names.